

PROPOSAL FOR DISCUSSION

Sadak Setu Digital Stack

Faster road project reports, smarter cost checks, paperless approvals for Jharkhand's 12,736 km road network

Prepared for: **Road Construction Department, Government of Jharkhand**

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THE SEVEN MODULES

- 01 Smart Cost Estimation** · AI-powered benchmarking
Every estimate compared against similar past projects before it reaches the approving officer.
- 02 Auto-DPR Drafting** · AI-assisted report generation
Complete DPR drafts in days instead of months, from survey data to a ready document.
- 03 Compliance Checker** · Automated standards verification
Every DPR checked against engineering rules before it leaves the division.
- 04 Cost Assurance** · Independent cost verification
Three independent checks that catch what a single threshold cannot.
- 05 Digital Approval Tracking** · Paperless approval with deadlines
Every DPR tracked from submission to sanction, with automatic escalation when delays occur.
- 06 Project Knowledge Base** · Searchable archive of all past projects
Every past DPR and sanction order, searchable in seconds: institutional memory that survives transfers.
- 07 GIS Road Map** · Interactive map of all roads and projects
One map linking every road to its project status, cost and approval stage.

Each module is set out in the same way: the problem, what we will build, how, the benefits and cost. Prices exclude GST and are valid for 90 days from the date of this proposal.

Portfolio at a glance

Jharkhand's Road Construction Department manages a 12,736 km network across 24 districts, with ₹6,601 crore for roads in 2026–27. Every road project begins with a Detailed Project Report (DPR): a document that today is drafted in varying formats, priced by hand, checked manually at several levels, moved as a physical file and never systematically compared with similar past projects. The seven modules below address each stage of this process, from cost estimation to final approval, and each one directly saves time, money or risk.

#	Module	What it solves	Status today	One-time	Recurring / yr
01	Smart Cost Estimation	No benchmarking; estimates vary widely for similar roads	Manual, no comparison system	₹31 lakh	₹5.5 lakh
02	Auto-DPR Drafting	Weeks of repetitive drafting; inconsistent formats	Manual process	₹40 lakh	₹8 lakh
03	Compliance Checker	DPRs returned for errors that could be caught early	Manual review only	₹18 lakh	₹3 lakh
04	Cost Assurance	No independent check that an estimate is reasonable	No system exists	₹9 lakh	₹2 lakh
05	Digital Approval Tracking	Physical file movement; no visibility into delays	File-based	₹23 lakh	₹4.5 lakh
06	Project Knowledge Base	Past DPRs not searchable; knowledge lost on transfer	Paper archives only	₹28.5 lakh	₹4.5 lakh
07	GIS Road Map	Road map not linked to DPR status and budget	GIS portal, not linked to DPRs	₹11.5 lakh	₹2.5 lakh
Whole portfolio				₹1.61 cr	₹30 lakh

For scale: the whole portfolio costs about 0.024% of RCD's ₹6,601 crore roads allocation for 2026–27.

→ HOW THE SEVEN FIT TOGETHER

One DPR, one digital thread. Smart Cost Estimation (01) benchmarks every project at inception. Auto-DPR Drafting (02) generates a complete report in days instead of months. The Compliance Checker (03) catches errors before the DPR leaves the division. Cost Assurance (04) independently verifies that the estimate is reasonable. Digital Approval Tracking (05) moves the DPR through a paperless approval chain with deadline enforcement. The Knowledge Base (06) makes every past project searchable. The GIS Road Map (07) places every project on RCD's map. Each can be built on its own, but they share a common platform and dashboard, so every module added makes the others more useful.

RECOMMENDED STARTING POINTS

01 Smart Cost Estimation:

the foundation. Digitise past DPRs, build the benchmark and deliver a cost comparison dashboard within months, giving immediate visibility to the Secretary and Engineer-in-Chief.

04 Cost Assurance:

builds on Module 01. Once benchmarking is live, adding independent cost verification needs minimal extra effort.

02 Auto-DPR Drafting:

the biggest time saver. Pilot with the 3 most common project types in Ranchi circle to demonstrate the reduction in drafting time.

Smart Cost Estimation

AI-powered benchmarking

Every estimate compared against similar past projects before it reaches the approving officer.

OWNER & PARTNER DEPARTMENTS RCD: Central Design Organisation (CDO); Chief Engineer (Communication); Finance; JAP-IT	STATUS TODAY Not started. Each division or DPR consultant prepares estimates manually. No system compares a new estimate with similar past sanctions. The last fully published RCD Schedule of Rates appears to be 2021–22, with later rates issued through circulars and tender notices.
TIMELINE Pilot 10–12 weeks · Full rollout 5–6 months	COST One-time ₹31 lakh · Recurring ₹5.5 lakh / yr

1 THE PROBLEM

- RCD has ₹6,601 crore for roads in 2026–27, yet every cost estimate is built from scratch with no comparison to what similar projects actually cost.
- Two similar roads in similar terrain can carry very different per-km estimates, with no system to flag or explain the gap.
- Rates come from the last published SoR, later circulars and tender notices, so the same item can be priced differently in two divisions.
- Review meetings rely on figures compiled by hand from paper files and spreadsheets across circles.

2 WHAT WE WILL BUILD

An AI system trained on RCD's past DPRs and sanction orders. Given a new project's basic details (road class, length, width, terrain, traffic, district), it generates a benchmark cost and shows how it compares with similar approved projects.

- **Plain-language explanations:** Shows exactly why a project costs more or less than comparable ones, and which items drive the difference.
- **Digital rate library:** Current SoR items, circulars and approved rate analyses in one versioned library, applied automatically.
- **Outlier alerts:** Estimates significantly above or below expectation are flagged with an explanation before they reach the approver.
- **Review dashboard:** A live view of estimates across circles, with comparison charts and a weekly review pack in Hindi and English.

WHY THIS IS DIFFERENT

Most estimation tools apply fixed rules or averages. This system learns from what RCD actually sanctioned and explains every figure in plain language. It also powers Module 04 (Cost Assurance), so independent cost verification needs little extra effort.

3 HOW WE WILL DO IT

Phase	Duration	What we do	Output
0. Discovery	3 weeks	Map DPR archives at Project Bhawan and circle offices; agree data access with CDO and Finance.	Baseline assessment
1. Pilot	8–10 weeks	Digitise the top 200 past DPRs; build the rate library and benchmark; pilot in Ranchi circle.	Live cost comparison dashboard
2. Rollout	3–4 months	All circles; weekly review pack; connect to Modules 04 and 06.	State-wide benchmarking
3. Run	Ongoing	Benchmark refreshed quarterly with new sanctions; monthly accuracy reports.	Monthly insights

Hosting: Jharkhand State Data Centre (JAP-IT), on the state's servers, with role-based access and a full audit trail.

4 BENEFITS

For the Government

- Every estimate compared with similar past projects before approval
- Evidence-based review for the Secretary and Engineer-in-Chief
- Target: trimming even 0.5% from inflated estimates saves about ₹30 crore a year on a ₹6,000 crore capital outlay
- Full audit trail for AG and vigilance review

For contractors, consultants & citizens

- Clear benchmarks for consultants and contractors
- Transparent cost basis reduces disputes during tendering
- Engineers freed from manual rate lookups for field supervision

5 COST

Component	Amount
Development: pilot (Phases 0–1)	₹11.5 lakh
Development: full rollout (Phase 2), incl. security audit	₹19.5 lakh
One-time total	₹31 lakh
Infrastructure (servers, storage, AI server, network)	<i>Provided by the State Data Centre</i>
Operations & maintenance per year (support, updates, helpdesk)	₹5.5 lakh
Recurring total per year	₹5.5 lakh

Includes deploying the platform on the State Data Centre's servers. Figures exclude GST.

6 KEY RISKS

Historical data quality varies	Start with the 200 highest-value recent DPRs; combine scanning with manual validation; accuracy improves as data grows.
Officers see it as surveillance	Positioned as decision support: it helps engineers strengthen their estimates and does not replace their judgement.
Rates change mid-year	The rate library is versioned; benchmarks recalibrate automatically when circulars change rates.

Auto-DPR Drafting

AI-assisted report generation

Complete DPR drafts in days instead of months, from survey data to a ready document.

OWNER & PARTNER DEPARTMENTS RCD: CDO; divisions; empanelled DPR consultants; JAP-IT	STATUS TODAY Not started. DPRs are drafted by divisions or empanelled consultants in varying formats. Much of the content repeats across similar projects, yet each is written from scratch.
TIMELINE Pilot 12 weeks · Full rollout 5–6 months (top 5 project types)	COST One-time ₹40 lakh · Recurring ₹8 lakh / yr

1 THE PROBLEM

- A typical DPR takes weeks of engineering time to assemble, much of it repetitive chapters and calculations.
- Divisions and consultants use different formats, so CDO reviewers spend time finding information before checking it, and DPRs come back for corrections.
- Pavement design is done in individual spreadsheets; one input error cascades through the estimate. Heavy coal and mineral traffic on some corridors makes this riskier.
- Forest-land, land-acquisition and utility-shifting sections are often incomplete, which later causes delays during execution.

2 WHAT WE WILL BUILD

Given survey data (alignment, soil conditions, traffic, drainage), the system generates a complete draft DPR in RCD's standard format, downloadable as a Word document:

- **Automated pavement design:** Layer thicknesses calculated to IRC:37, including heavy-axle mining traffic, with no manual spreadsheet work.
- **Complete Bill of Quantities:** Items linked to the Module 01 rate library with automatic rate application.
- **AI-written narrative:** Executive summary, justification and site description drafted by an AI model running inside the State Data Centre; engineers review and edit instead of writing from scratch.
- **Clearance chapter:** Structured sections for forest land, land acquisition and utilities, so gaps are visible at draft stage.
- **One-click Word export:** Standard RCD format, checked against Module 01 benchmarks before export.

WHY THIS IS DIFFERENT

This is not a template filler. The system designs the pavement from IRC rules, prices it from RCD rates, and uses AI only to write the narrative. The engineer's role shifts from document assembly to engineering review.

3 HOW WE WILL DO IT

Phase	Duration	What we do	Output
0. Baseline	3 weeks	Collect the top 5 DPR types by volume; agree the standard format with CDO.	DPR template library
1. Pilot	10–12 weeks	Build the design engine, BoQ generator and AI narrative; pilot in Ranchi circle on 3 project types.	10–15 AI-drafted DPRs
2. Rollout	4–5 months	All 5 project types, all circles and consultants; connect to Modules 03 and 04; training workshops.	State-wide DPR automation
3. Run	Ongoing	Template updates, rate synchronisation, quarterly accuracy review.	Continuous improvement

Output: Bilingual (Hindi and English). The AI model is an open-source model running on the State Data Centre's own GPU server; no data leaves the data centre.

4 BENEFITS

For the Government

- DPR drafting time for common project types cut from weeks to days
- Consistent format and calculations across all 24 districts and all consultants
- Senior engineering time freed for field supervision instead of paperwork
- Fewer return-for-correction rounds with built-in checks

For contractors, consultants & citizens

- Consultants receive clear, standardised DPR templates
- Faster DPR-to-tender pipeline gets projects to construction sooner
- Standardised estimates reduce tendering disputes

5 COST

Component	Amount
Development: pilot (Phases 0–1)	₹12.5 lakh
Development: full rollout (Phase 2), incl. security audit	₹27.5 lakh
One-time total	₹40 lakh
Infrastructure (servers, storage, AI server, network)	<i>Provided by the State Data Centre</i>
Operations & maintenance per year (support, updates, helpdesk)	₹8 lakh
Recurring total per year	₹8 lakh

Includes installing and tuning the in-house AI model on the State Data Centre's GPU server. Figures exclude GST.

6 KEY RISKS

Engineers resist AI-drafted DPRs	AI drafts only the narrative; design and costs are rule-based. The engineer reviews and signs, and every edit is tracked.
Engineering standards change	Design rules are version-controlled; new IRC editions are coded within weeks of notification.
Site-specific conditions not captured	Site observations, photographs and local conditions are always added by the engineer.

Compliance Checker

Automated standards verification

Every DPR checked against engineering rules before it leaves the division.

OWNER & PARTNER DEPARTMENTS RCD: CDO; Quality Control wing; divisions	STATUS TODAY Manual review only. Errors are caught late, at circle or CDO level, after review time has already been spent at earlier stages. No automated checking exists.
TIMELINE Pilot 8–10 weeks · Full rollout 3–4 months	COST One-time ₹18 lakh · Recurring ₹3 lakh / yr

1 THE PROBLEM

- DPRs are reviewed by hand at each level (Division → Circle → CDO). The same basic errors are caught late, wasting review time at every level.
- The most common return reasons (pavement thickness, carriageway width for the road class, drainage sizing, missing chapters) are mechanically checkable.
- Alignments through forest land or sensitive areas are often identified late, when clearance delays are hardest to absorb.
- No standard digital checklist exists, so each reviewer applies their own interpretation.

2 WHAT WE WILL BUILD

An automated checker that verifies every DPR against IRC standards and RCD norms before it leaves the division:

- **Pavement thickness:** Road layers meet minimum standards for the expected traffic.
- **Road width:** Carriageway and formation width match the road's classification.
- **Drainage sizing:** Drains and culverts are correctly sized for the catchment.
- **Clearance flags:** Alignments crossing forest land are flagged early for the clearance process.
- **Complete documentation:** All required chapters, drawings and annexures are present before the file can move forward.

WHY THIS IS DIFFERENT

This is not a checklist on paper. Each rule runs automatically, cites the standard and clause it checks, and tells the officer exactly what failed and why.

3 HOW WE WILL DO IT

Phase	Duration	What we do	Output
0. Baseline	3 weeks	List the top 15 return reasons across circles; agree the rules with CDO.	Rule catalogue
1. Pilot	8–10 weeks	Automate the first rules; run alongside manual review to validate.	10 rules automated
2. Rollout	2–3 months	30+ rules; connect to Modules 02 and 05; training workshops.	State-wide checker
3. Run	Ongoing	New rules added when standards are updated.	Always-current rules

Approach: Each rule cites the standard, clause and threshold. Runs in shadow mode before going live.

4 BENEFITS

For the Government

- Return cycles for checkable errors reduced to near zero
- Reviewers freed to focus on engineering judgement
- Full audit trail protects officers and supports accountability
- Clearance issues surfaced at the earliest stage

For contractors, consultants & citizens

- DPRs that pass move faster, with predictable timelines
- Consultants get instant feedback on gaps
- Clear compliance report for every DPR

5 COST

Component	Amount
Development: pilot (Phases 0–1)	₹6.5 lakh
Development: full rollout (Phase 2), incl. security audit	₹11.5 lakh
One-time total	₹18 lakh
Infrastructure (servers, storage, AI server, network)	<i>Provided by the State Data Centre</i>
Operations & maintenance per year (support, updates, helpdesk)	₹3 lakh
Recurring total per year	₹3 lakh

Figures exclude GST.

6 KEY RISKS

Rules too rigid for field conditions	Each rule has a review band, not just pass/fail. Engineers can override with a documented reason.
Standards change faster than the system	Rule updates are coded within two weeks of notification under maintenance.
Officers see it as replacing judgement	Shadow mode for one cycle before it becomes mandatory.

Cost Assurance

Independent cost verification

Three independent checks that catch what a single threshold cannot.

OWNER & PARTNER DEPARTMENTS RCD: CDO; Finance; Vigilance	STATUS TODAY Not started. An estimate is checked mainly against its own build-up, which is a circular check. No independent benchmark or peer comparison exists.
TIMELINE Pilot 6–8 weeks (builds on Module 01) · Full rollout 2–3 months	COST One-time ₹9 lakh · Recurring ₹2 lakh / yr

1 THE PROBLEM

- A fixed-percentage check cannot tell genuine cost variation (difficult terrain, mining traffic) from estimate inflation.
- The item-wise total and an independent benchmark are never compared today.
- There is no way to compare a project's cost per km with similar sanctioned projects in the same region.
- When an engineer changes a system-generated figure, the change is not tracked.

2 WHAT WE WILL BUILD

A verification layer that cross-checks each estimate from three angles and attaches an assurance report to every DPR before approval:

- **Check 1: item-wise vs benchmark:** The detailed cost total is compared with the Module 01 benchmark; significant gaps are flagged with an explanation.
- **Check 2: manual override detection:** Changes to system-generated figures are flagged; small changes get a notice, large ones a warning.
- **Check 3: peer comparison:** Cost per km compared with similar sanctioned projects of the same road class and region.

WHY THIS IS DIFFERENT

Three independent checks, each looking at cost from a different angle, together provide an assurance layer that no manual review can match at scale.

3 HOW WE WILL DO IT

Phase	Duration	What we do	Output
0. Data prep	2 weeks	Requires Module 01. Assemble a database of sanctioned projects with cost per km.	Peer database
1. Pilot	6–8 weeks	Build the three checks; attach reports to pilot DPRs; validate against manual review.	Assurance reports on pilot DPRs
2. Rollout	1–2 months	All circles; connect to Modules 03 and 05; dashboard for Finance.	State-wide cost assurance
3. Run	Ongoing	Thresholds recalibrated quarterly; anomaly report for the Secretary.	Quarterly assurance report

Approach: A verification layer on the Module 01 infrastructure; no separate system is required.

4 BENEFITS

For the Government

- Independent, evidence-based cost verification on every DPR
- Vigilance-ready trail of every check, threshold and override
- Target: every DPR with a cost anomaly flagged before it reaches the Secretary
- Accuracy improves automatically as more projects are sanctioned

For contractors, consultants & citizens

- Engineers see where their estimate stands against peers
- Transparent benchmarking builds trust in estimates
- Fewer post-sanction disputes and revisions

5 COST

Component	Amount
Development: pilot (Phases 0–1)	₹4 lakh
Development: full rollout (Phase 2), incl. security audit	₹5 lakh
One-time total	₹9 lakh
Infrastructure (servers, storage, AI server, network)	<i>Provided by the State Data Centre</i>
Operations & maintenance per year (support, updates, helpdesk)	₹2 lakh
Recurring total per year	₹2 lakh

Builds on Module 01, so no separate system is needed. Figures exclude GST.

6 KEY RISKS

Too many false alerts initially	Shadow mode for one cycle; calibrate against known-good sanctions.
Finance sees it as encroachment	Designed jointly with Finance from Phase 0; the report supports their review and does not replace it.
Too few comparable projects	The report says so explicitly rather than making a misleading comparison.

Digital Approval Tracking

Paperless approval with deadlines

Every DPR tracked from submission to sanction, with automatic escalation when delays occur.

OWNER & PARTNER DEPARTMENTS RCD: all divisions and circles; CDO; Engineer-in-Chief; Department Secretariat; Finance; JAP-IT	STATUS TODAY File-based. DPR files move Division → Circle → CDO → Engineer-in-Chief → Department → Finance. No DPR-specific system shows which file is where or how long it has been there (to be confirmed with the Department).
TIMELINE Pilot 10–12 weeks · Full rollout 4–5 months	COST One-time ₹23 lakh · Recurring ₹4.5 lakh / yr

1 THE PROBLEM

- File movement through several approval stages takes months, and nobody can see the full pipeline at a glance.
- No stage deadlines are enforced; a file can wait for weeks without escalation.
- Returned DPRs carry handwritten remarks, and submitters often cannot tell exactly what to fix.
- Many roles, from Junior Engineer to Secretary, act on DPRs, but there is no role-based view or priority queue.

2 WHAT WE WILL BUILD

A digital approval system that moves every DPR through a defined chain (Draft → Division → Circle → CDO → Engineer-in-Chief → Department → Sanctioned) with clear deadlines and automatic escalation. It complements, and can link to, the state's e-Office:

- **Role-based dashboards:** Each officer sees only the DPRs needing action, with countdown timers and overdue items flagged.
- **Deadline enforcement:** Agreed days per stage, set from observed data, with automatic escalation at 80% and 100% of the time allowed.
- **Priority queuing:** The most urgent and highest-value items surface first.
- **Clear return feedback:** Specific sections needing correction are highlighted, with no ambiguity.

WHY THIS IS DIFFERENT

This is not e-Office with a DPR label. Stages cannot be skipped, deadlines escalate automatically, and a DPR cannot enter the chain until its compliance and cost-assurance reports (Modules 03–04) are attached.

3 HOW WE WILL DO IT

Phase	Duration	What we do	Output
0. Map	3 weeks	Document the actual approval chain in 2 circles; identify bottlenecks; agree deadline targets with the Engineer-in-Chief.	Process map and deadlines
1. Pilot	10–12 weeks	Build the approval system and dashboards; pilot in Ranchi circle; connect to Modules 02–04.	Live digital approval for one circle
2. Rollout	3–4 months	All circles; mobile and SMS alerts; Secretary's dashboard; training for all roles.	State-wide digital approvals
3. Run	Ongoing	Deadline tuning, bottleneck analytics, quarterly reports.	Monthly pipeline analytics

Approach: Web-based with SMS and e-mail alerts; offline drafts sync when connectivity returns.

4 BENEFITS

For the Government

- Real-time view of every DPR for the Secretary and Engineer-in-Chief
- Approval time targeted to fall from months to weeks
- Clear return feedback reduces back-and-forth
- Analytics show which stage, office and project type cause delays

For contractors, consultants & citizens

- Submitters see status in real time, with no phone calls
- Clear feedback means faster correction and resubmission
- A digital trail replaces physical file tracking

5 COST

Component	Amount
Development: pilot (Phases 0–1)	₹9 lakh
Development: full rollout (Phase 2), incl. security audit	₹14 lakh
One-time total	₹23 lakh
Infrastructure (servers, storage, AI server, network)	<i>Provided by the State Data Centre</i>
Operations & maintenance per year (support, updates, helpdesk)	₹4.5 lakh
Recurring total per year	₹4.5 lakh

Deadline alerts use the state's SMS and e-mail gateways. Figures exclude GST.

6 KEY RISKS

Officers bypass the system	Final sanction requires the digital record, so upstream adoption follows.
Poor connectivity in field offices	Light web app, offline drafts and SMS fallback for alerts.
Deadlines are unrealistic	Set from observed turnaround times in Phase 0 and tuned quarterly.

Project Knowledge Base

Searchable archive of all past projects

Every past DPR and sanction order, searchable in seconds: institutional memory that survives transfers.

OWNER & PARTNER DEPARTMENTS RCD: CDO; divisions; record rooms at Project Bhawan and circle offices	STATUS TODAY Paper archives only. DPRs and sanction orders from the 25 years since the state's formation sit in record rooms. They cannot be searched or compared.
TIMELINE Pilot 8–10 weeks · Full rollout 3–4 months	COST One-time ₹28.5 lakh · Recurring ₹4.5 lakh / yr

1 THE PROBLEM

- Institutional memory leaves with officers on transfer; the engineer who designed a similar road is now elsewhere, and the DPR is in a cabinet.
- An engineer preparing a new DPR cannot quickly find similar past projects for reference or cost comparison.
- The cost benchmarking in Modules 01 and 04 needs historical data that is not yet in searchable form.

2 WHAT WE WILL BUILD

A digital archive of past DPRs and sanction orders, with AI search that understands content, not just keywords:

- **Smart search:** "Two-lane road with a minor bridge in hilly forest terrain" finds relevant projects even when the words differ.
- **Cost comparison:** For any new project, the cost per km of the five most similar sanctioned projects.
- **Reference library:** Complete past DPRs one click away while drafting.
- **Data foundation:** Feeds Modules 01 and 04 with the historical data they need.

WHY THIS IS DIFFERENT

This is not a filing system. The AI understands what each project is about, which makes similarity search and cost comparison possible.

3 HOW WE WILL DO IT

Phase	Duration	What we do	Output
0. Inventory	3 weeks	Survey record rooms; prioritise recent, high-value and common project types.	Digitisation plan
1. Pilot	8–10 weeks	Scan and index the first 300 DPRs; build search; pilot with engineers in Ranchi circle.	Searchable archive of 300 DPRs
2. Rollout	2–3 months	Up to ~800 DPRs; all circles; connect to Modules 01, 02 and 04.	State-wide knowledge base
3. Run	Ongoing	Every newly sanctioned DPR added automatically.	Self-growing archive

Approach: Paper DPRs scanned and digitised with OCR; AI search and similarity matching run on models hosted inside the State Data Centre.

4 BENEFITS

For the Government

- Institutional memory preserved through transfers
- Comparable projects found in seconds instead of days
- Feeds Modules 01 and 04, improving accuracy over time
- A structured view of the Department's project history

For contractors, consultants & citizens

- Consultants can calibrate estimates against approved precedent
- Transparent record reduces disputes about past approvals

5 COST

Component	Amount
Development: pilot (Phases 0–1)	₹14 lakh
Development: full rollout (Phase 2), incl. security audit	₹14.5 lakh
One-time total	₹28.5 lakh
Infrastructure (servers, storage, AI server, network)	<i>Provided by the State Data Centre</i>
Operations & maintenance per year (support, updates, helpdesk)	₹4.5 lakh
Recurring total per year	₹4.5 lakh

Includes scanning and OCR of up to ~800 past DPRs. Figures exclude GST.

6 KEY RISKS

Archive condition is poor	Start with digital and recent files; manual correction where OCR fails.
Search results are imprecise	Combined AI and keyword search; engineer feedback improves ranking.
Circles reluctant to share	Read-only central access; circles retain ownership of their records.

GIS Road Map

Interactive map of all roads and projects

One map linking every road to its project status, cost and approval stage.

OWNER & PARTNER DEPARTMENTS

RCD GIS cell; CDO; Forest, Environment & Climate Change Department (layers); JAP-IT

STATUS TODAY

GIS portal exists, not linked to DPRs. RCD already runs a GIS portal for State Highways and Major District Roads. It does not show DPRs in preparation or under approval.

TIMELINE

Pilot 6–8 weeks · Full rollout 2 months

COST

One-time ₹11.5 lakh · Recurring ₹2.5 lakh / yr

1 THE PROBLEM

- RCD's 12,736 km network has no single view linking roads to DPRs in preparation, approval stage and budget.
- Planning decisions (which roads to prioritise, where to allocate budget) are made from tables and lists.
- Overlaps, where two proposals target the same stretch, and conflicts with forest land or elephant corridors are caught late.

2 WHAT WE WILL BUILD

An extension of RCD's existing GIS that shows every road segment, active DPR, approval status and cost on one screen:

- **Project overlay:** Proposed, under-approval and sanctioned DPRs on their road segments; click any road to see details.
- **Filtering:** By circle, district, road class or project status; the Secretary sees the state, the Superintending Engineer their circle.
- **Overlap detection:** Automatic alert when two proposals target the same stretch.
- **Sensitivity overlay:** Forest land and wildlife-corridor layers, where shared, to flag clearance needs at DPR stage.

WHY THIS IS DIFFERENT

This is not another map viewer. It builds on the GIS the state already has and connects it to the live DPR pipeline, turning it into a planning tool.

3 HOW WE WILL DO IT

Phase	Duration	What we do	Output
0. Data prep	2 weeks	Connect to the RCD GIS portal; locate existing DPRs on road segments.	Road–DPR linked dataset
1. Pilot	6–8 weeks	Project and approval layers; connect to Module 05 for live status; pilot in Ranchi circle.	Linked map for one circle
2. Rollout	1–2 months	All circles; overlap detection; sensitivity overlay; budget heat map.	State-wide planning map
3. Run	Ongoing	New DPRs appear automatically; layers refreshed with the GIS cell.	Living infrastructure map

Approach: Built on the existing RCD GIS portal and open GIS standards, integrated with JAP-IT.

4 BENEFITS

For the Government

- One visual view of the network linked to project status
- Overlap detection prevents duplicate spending on the same stretch
- Forest and corridor conflicts caught before DPRs are finalised
- Budget allocation visible by district

For contractors, consultants & citizens

- Contractors see where work is active and upcoming
- Citizens can see which roads in their district are being improved

5 COST

Component	Amount
Development: pilot (Phases 0–1)	₹4 lakh
Development: full rollout (Phase 2), incl. security audit	₹7.5 lakh
One-time total	₹11.5 lakh
Infrastructure (servers, storage, AI server, network)	<i>Provided by the State Data Centre</i>
Operations & maintenance per year (support, updates, helpdesk)	₹2.5 lakh
Recurring total per year	₹2.5 lakh

Reuses the existing RCD GIS portal, so no new map platform is purchased. Figures exclude GST.

6 KEY RISKS

Road data is incomplete	Start with State Highways and Major District Roads (best data); gaps are flagged, not hidden.
Limited access to the GIS portal	Integration agreed with the GIS cell and JAP-IT in week one; fallback is a lightweight layer on open base maps.
Map data becomes outdated	New DPRs update the map automatically; regular refresh with the GIS cell.

How we will work with the Department

Start small, prove value	A 6–8 week proof of concept using the Department's own data. Scale up only after results are visible.
Transparent procurement	Engagement through GeM, JAP-IT or an empanelled system integrator. Rollouts through open tender.
Data stays with the state	Everything runs in the Jharkhand State Data Centre (JAP-IT), including the AI models. No data goes to any external cloud or AI service. Compliant with the DPDP Act 2023.
Security first	Independent security audit before every go-live. Role-based access and tamper-proof audit logs.
No vendor lock-in	Open-source technology, source code in escrow, full documentation and training for state officers.
Hindi and English	All screens, reports and training material in Hindi and English.
Pay for outcomes	Every contract carries measurable targets (approval days, DPRs generated, cost accuracy) with monthly reports.
Built for Jharkhand	Delivery team based in Kolkata with on-site presence in Ranchi, and internships for students from Jharkhand engineering colleges.

i COST ASSUMPTIONS

- Prices cover design, development, integration, project management, on-site support in Ranchi, training, and contingency for data-quality and integration work.
- Prices assume the seven modules are delivered together on one shared platform; modules bought separately may cost more.
- Pilot = initial proof of concept; Rollout = state-wide deployment. Annual maintenance is typically 15–20% of build cost.
- All infrastructure is provided by the Jharkhand State Data Centre (JAP-IT): servers, storage, GPU server for the AI models, network, security, backup and disaster recovery (see the hosting requirements below). Kirnova does not supply or charge for infrastructure.
- All data and AI models stay inside the State Data Centre. The AI models are open-source models deployed on the state's servers; no data is sent to any external cloud or AI service.
- All amounts exclude GST and are valid for 90 days from the date of this proposal. They assume the scope described here; changes in scope are priced separately.

ii HOSTING REQUIREMENTS (PROVIDED BY THE STATE DATA CENTRE)

Sized for the full rollout of all seven modules; final sizing is confirmed with JAP-IT in Phase 0.

Application servers	2 virtual machines, 8 vCPU / 32 GB RAM each, load-balanced
Database	2 virtual machines, 8 vCPU / 32 GB RAM, 1 TB SSD each (PostgreSQL with PostGIS, primary and standby)
Search and document services	1 virtual machine, 8 vCPU / 32 GB RAM, 500 GB SSD
AI server	1 GPU server with one NVIDIA L4 / A10 (24 GB) or better, 16 vCPU / 64 GB RAM, for the in-house language and search models
Storage	2 TB for DPR documents and scanned archives, plus backup as per the data centre's policy
Staging environment	2 virtual machines, 8 vCPU / 32 GB RAM
Network and services	SSL certificates, access to the state SMS and e-mail gateways, connectivity to RCD offices over the state network
Backup and disaster recovery	As per the State Data Centre's standard policy

iii SOURCES (PUBLIC INFORMATION, ACCESSED SEPTEMBER 2026)

1. RCD Jharkhand: road network achievements (12,736 km; SH 1,231.9 km; MDR 4,845.7 km) — <https://rcd.jharkhand.gov.in/achivements.php>
2. RCD Jharkhand: organisation (Engineer-in-Chief; CE, Central Design Organisation) — <https://rcd.jharkhand.gov.in/organisation.php>
3. RCD GIS portal — https://gis.jharkhand.gov.in/rcd_portal/index.php
4. Jharkhand Budget 2026–27 (Drishti IAS): RCD ₹6,601.28 crore — <https://www.drishtiias.com/state-pcs-current-affairs/jharkhand-budget-202627>
5. Jharkhand Budget Analysis 2026–27 (PRS India) — <https://prsindia.org/budgets/states/jharkhand-budget-analysis-2026-27>
6. Jharkhand Budget 2026–27 overview (Elets eGov) — <https://egov.eletsonline.com/2026/02/jharkhand-budget-2026-27-unveils-%E2%82%B91-58-lakh-crore-plan-focused-on-welfare-infrastructure-and-economic-growth/>
7. Jharkhand RCD Schedule of Rates status (CivilShape) — <https://www.civilshape.in/2025/06/Jharkhand-Road-Construction-Department-SOR-Updated-in-2025-Jharkhand-RCD-SOR.html>
8. State-wise Schedule of Rates 2026 (InfraLens) — <https://infralens.in/knowledge/state-wise-sor-india-2026>
9. Jharkhand State Roads Project: forest land along project roads (ADB) — <https://www.adb.org/sites/default/files/project-documents/40005-ind-siee.pdf>
10. JAP-IT: nodal e-governance agency, State Data Centre — <https://japit.jharkhand.gov.in/>

Status descriptions are based on public reports up to September 2026 and should be confirmed with the Department before formal submission.